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Industrial Internship Report on

"Prediction of Agriculture Crop Production and Forecasting of Smart City Traffic Patterns"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship offered by upskill Campus and The IoT Academy in collaboration with the Industrial Partner, UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project, including the report, in 6 weeks. My project was My projects were Prediction of Agriculture Crop Production in India and Forecasting of Smart City Traffic Patterns. This internship gave me a very good opportunity to gain exposure to Industrial problems and to design/implement solutions for them. It was an overall great experience to have this internship.

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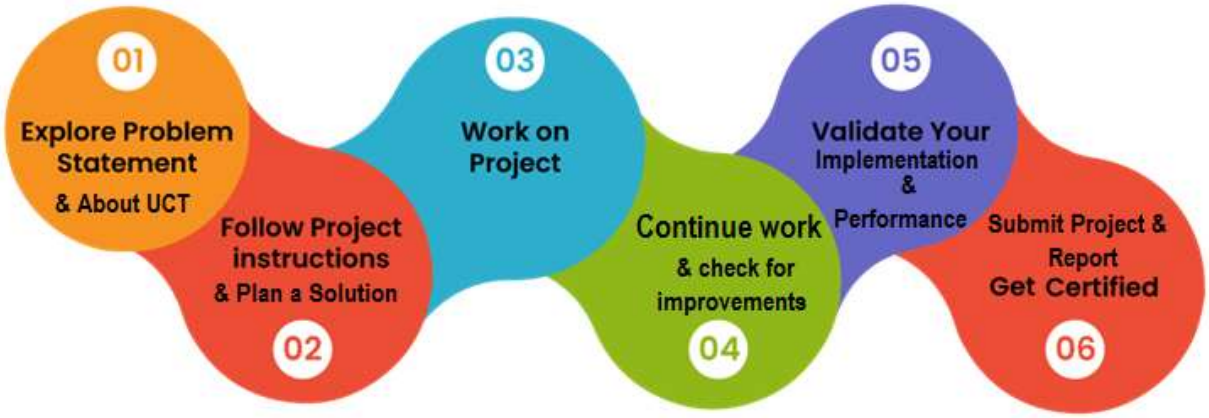
1 Preface

Summary of the whole 6 weeks' work.

This internship program was designed to provide practical exposure to real-world problems using machine learning techniques. Over the span of six weeks, I worked on agriculture and smart city-based projects.

The agriculture project focused on predicting crop production, while the smart city project focused on analyzing and forecasting traffic patterns. These projects helped me understand data preprocessing, visualization, and model implementation.

I would like to thank upSkill Campus, The IoT Academy, and UCT for providing this opportunity. I also extend my gratitude to mentors and peers who supported me throughout the internship.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

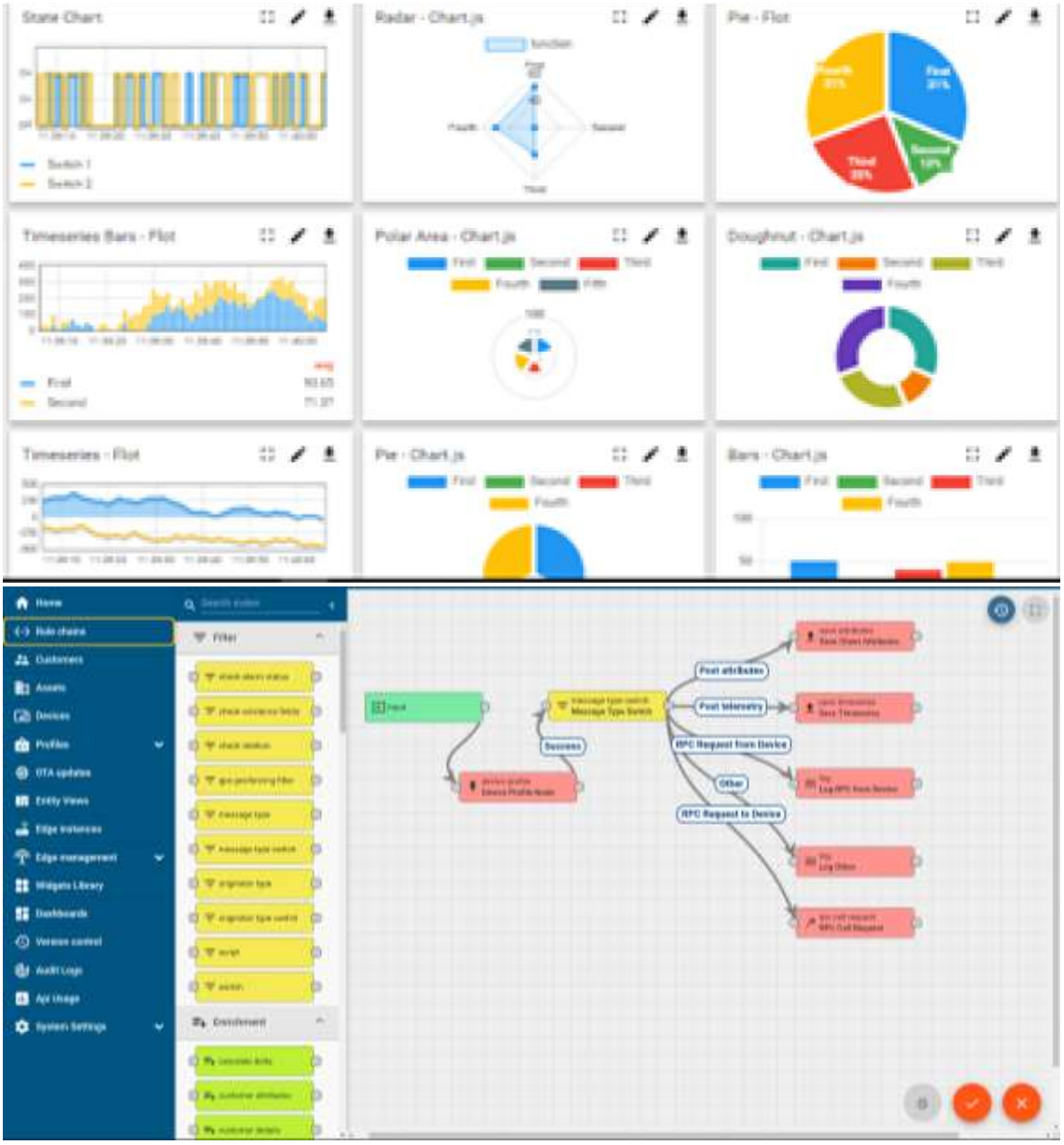
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Demo Org
Demo
CNC_S7_B1
CNC_S7_B2
CNC_S7_B4
Utilization
Utilization
Utilization

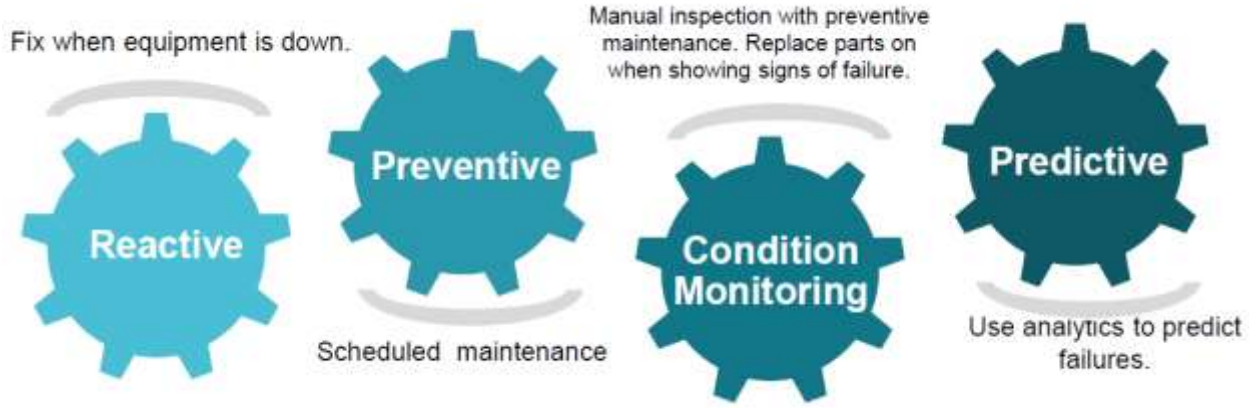
3D View
12:30
Stopped
Workplace Type
194724_27462-100-A3
Availability
Performance
Quality
OEE
Monitor

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

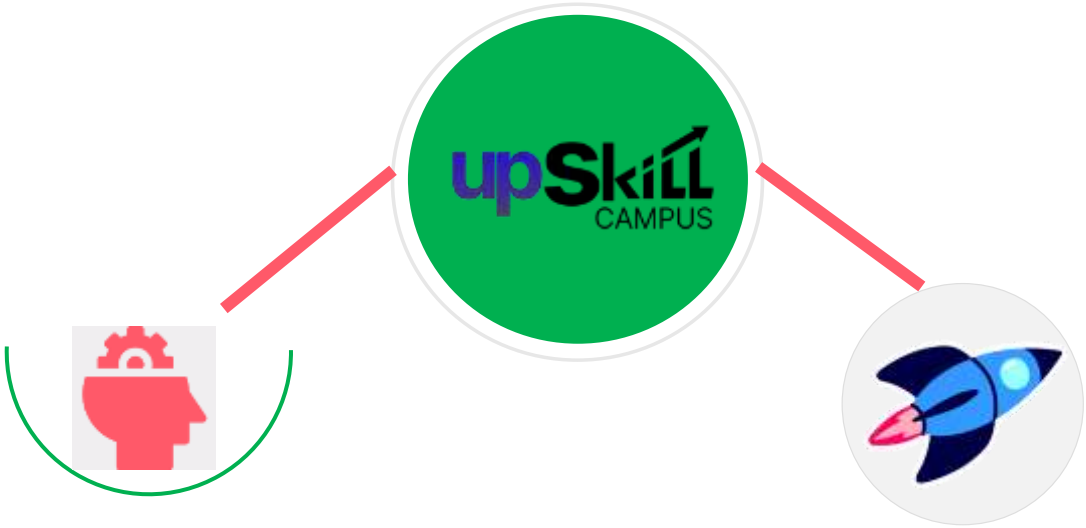
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

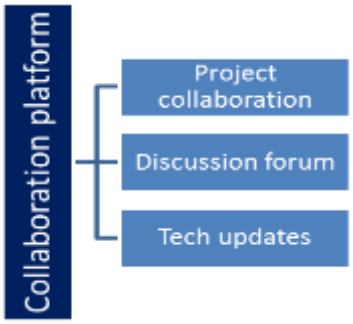
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

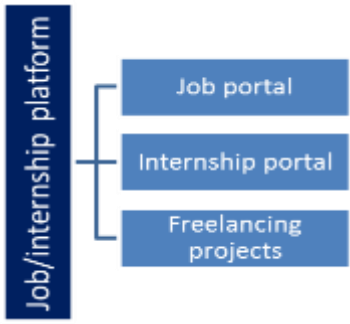


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/





2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1]

[2]

[3]

2.6 Glossary

Terms	Acronym

3 Problem Statement

In the assigned problem statement

Project 1: Agriculture Crop Production

Agriculture is one of the most important sectors in India. However, predicting crop production is challenging due to multiple influencing factors such as season, location, and cost. The problem is to build a model that can predict crop production accurately.

Project 2: Smart City Traffic Forecasting

Traffic congestion is a major issue in smart cities. The problem is to analyze traffic data and forecast traffic patterns at different junctions to improve urban planning and reduce congestion.

4 Existing and Proposed solution

Existing Solutions:

Traditional methods rely on manual analysis and lack accuracy.

Traditional methods rely on manual analysis

Lack of real-time prediction

Limited use of data-driven techniques

Proposed Solution:

Machine learning models are used to analyze data and generate predictions.

Use **machine learning models** for prediction

Apply **data preprocessing and visualization techniques**

Generate accurate predictions for decision-making

Value Addition:

Improved accuracy, faster predictions, and data-driven decision-making.

Improved accuracy

Faster decision-making

Data-driven insights

4.1 Code submission (Github link)

https://github.com/TrishaRajkumar/upskillCampus

Report submission (Github link):

https://github.com/TrishaRajkumar/upskillCampus

5 Proposed Design/ Model

The project follows these steps:

1. Data Collection

2. Data Preprocessing

3. Exploratory Data Analysis

4. Feature Selection

5. Model Training

6. Prediction

7. Evaluation

Models used

Linear Regression (Agriculture)

Random Forest (Traffic Prediction)

5.1 High-Level Diagram (if applicable)

```
+------------------+
|  Dataset Input  |
+------------------+
        ↓
+----------------------+
| Data Preprocessing   |
+----------------------+
        ↓
+----------------------+
| Feature Selection    |
+----------------------+
        ↓
+----------------------+
| Machine Learning     |
| Model Training       |
+----------------------+
        ↓
+----------------------+
| Prediction Output    |
+----------------------+
```

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low-Level Diagram (if applicable)

```
+------------------+
| Dataset (CSV)    |
+------------------+
        ↓
+----------------------+
| Handle Missing Data  |
+----------------------+
        ↓
+----------------------+
| Encoding (Categorical|
| → Numerical)         |
+----------------------+
        ↓
+----------------------+
| Train-Test Split     |
+----------------------+
        ↓
+-----------------------------+
| Model Training              |
| (Linear Regression / RF)    |
+-----------------------------+
        ↓
+----------------------+
```

```
| Prediction        |
+---------------------+
        ↓
+---------------------+
| Evaluation (MSE/MAE) |
+---------------------+
```

5.3 Interfaces (if applicable)

The system consists of the following interfaces:

Input Interface: CSV files such as agriculture.csv and traffic.csv are used as input datasets.

Processing Interface: Python environment using libraries like Pandas, NumPy, Scikit-learn for data processing and model training.

Output Interface: Prediction results and visualizations are displayed using Matplotlib graphs.

Block diagrams and flowcharts are used to represent the workflow of the system.

6 Performance Test

6.1 Test Plan/ Test Cases

Test Plan:

The dataset was divided into training and testing sets.

Split data into training and testing sets

Train the model on the training data

Evaluate using test data

6.2 Test Procedure

Test Procedure:

Models were trained, and predictions were made.

Mean Squared Error (Agriculture)

Mean Absolute Error (Traffic)

6.3 Performance Outcome

Performance Outcome:

Agriculture model evaluated using Mean Squared Error.

Traffic model evaluated using Mean Absolute Error.

Models produced reasonable predictions

Performance improved after preprocessing

7 My learnings

Learned data preprocessing, machine learning models, and real-world applications in agriculture and smart city domains. Improved Python and data analysis skills.

Gained hands-on experience in **machine learning projects**

Learned **data preprocessing and visualization techniques**

Improved skills in **Python, Pandas, and ML models**

Understood real-world applications of agriculture and smart city systems

8 Future work scope

Future work includes using advanced machine learning models, improving accuracy, and working with real-time datasets.

Use advanced models like **Deep Learning**

Work with **larger real-time datasets**

Improve prediction accuracy

Integrate with IoT-based smart systems